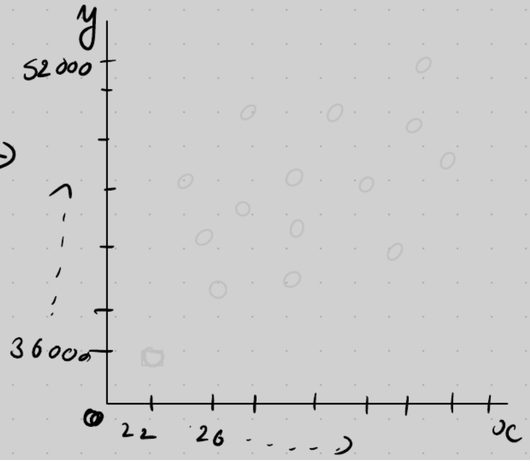
Made with Goodnotes

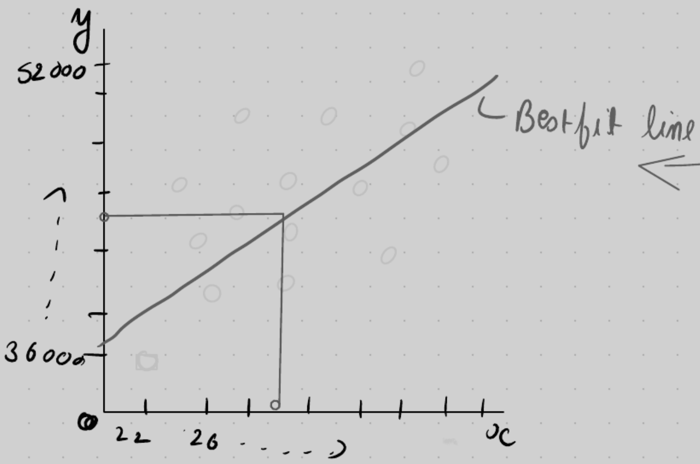
Linear Regression

Problem statement

Input ← | → output

Age	Salary
22	36000
26	38000
28	40000
34	48000
39	52000
42	⋮
46	
⋮	





Now lets see how you actually create a best fit line

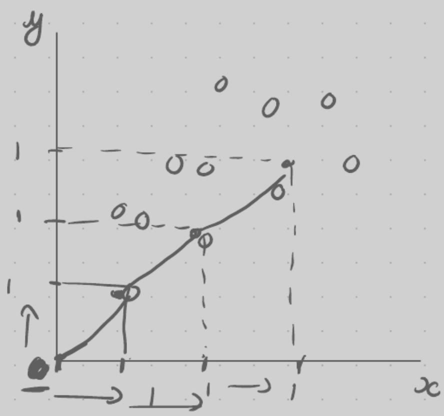
$y = mx + b$

$1 = m \rightarrow$ slope

$0 = b \Rightarrow$ intercept

$x \Rightarrow$ Data point

$y \Rightarrow$ Prediction



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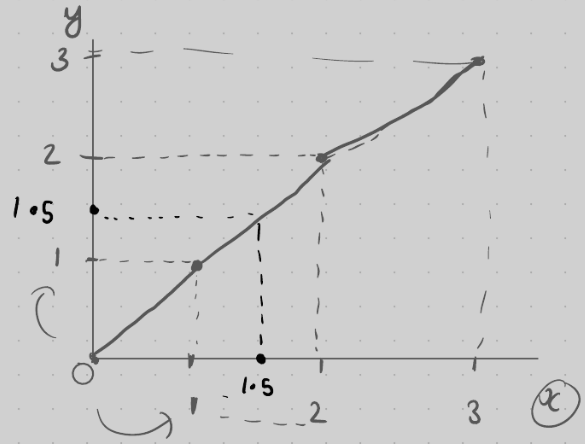
slope ← $m = 1$

intercept ← $b = 0$



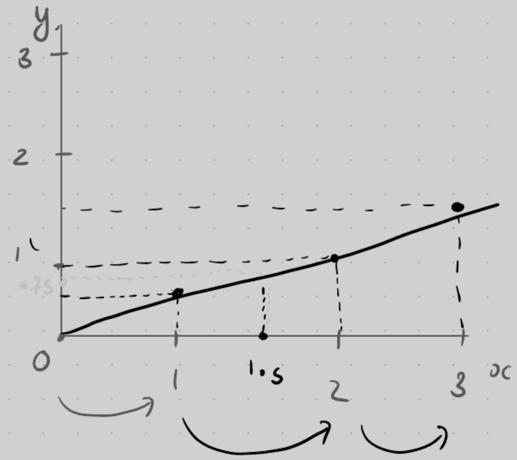
$y = x = 1.5$

$y = mx + b$

$y = 1(1.5) + 0 \quad = \quad y = 1.5$

$m = .5$

$b = 0$



$\overset{x}{1.5} = ?\, y$

$y = mx + b$

$y = .5(1.5) + 0$

$y = .75$

$h_\theta(\underline{x}) = \theta_0 + \theta_1 x$

$\theta_0 \Rightarrow$ intercept

$\theta_1 \Rightarrow$ slope

We have to find the perfect value of θ_0 and θ_1.

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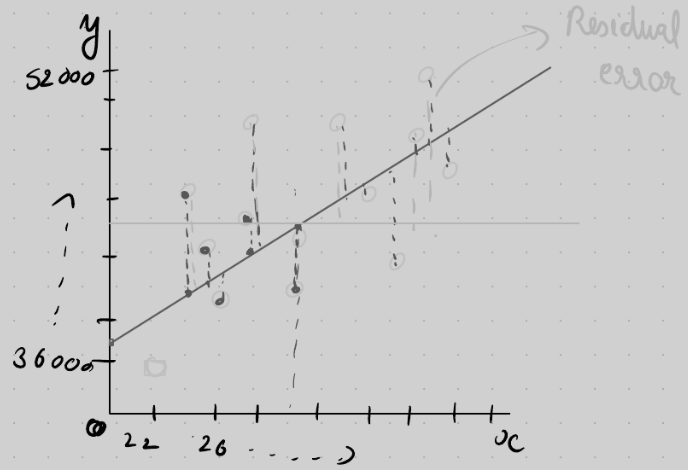
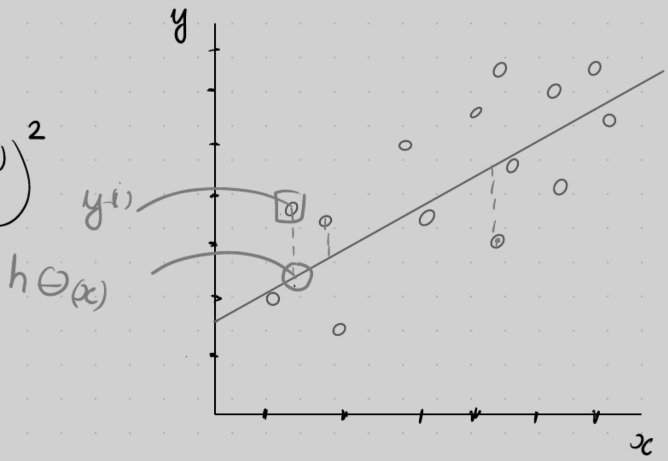


Cost function



$$J(\theta_0, \theta_1) = \sum_{i=1}^{m} \frac{1}{2m}\left(h\theta(x)^{(i)} - y^{(i)}\right)^2$$

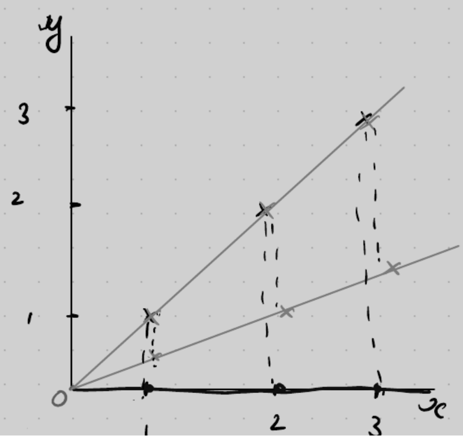
$\Downarrow$

cost function

You have to minimize the cost function by changing the value of θ_0, θ_1.

Gradient Descent

x	y
{ $\boxed{1}$,	1 }
{ 2,	2 }
{ 3,	3 }



$\theta_0 = 0$, $\theta_1 = \boxed{1}$

$(h\theta) = \theta_0 + \theta_1 x$

$h\theta) = 0 + 1(1) = 1$

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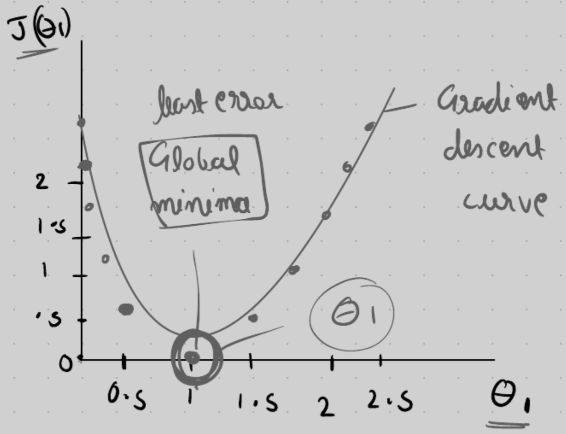
$h(\theta) = 0 + 1(2) = 2$

$h(\theta) = 0 + 1(3) = 3$



$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} \left(h_\theta(x)^{(i)} - y^{(i)} \right)^2$$

$$= \frac{1}{6} \left[(1-1)^2 + (2-2)^2 + (3-3)^2 \right]$$

$$= \frac{1}{6} (0^2 + 0^2 + 0^2)$$

$$= 0 \qquad J(\theta_1) = 0$$

$\theta_1 = .5, \theta_0 = 0$

$(h\theta) = \theta_0 + \theta_1 x_1$

$$J(\theta_1) = \frac{1}{6} \left[(1-0.5)^2 + (1-2)^2 + (3-1.5)^2 \right] =$$

$0 + .5(1) = .5$

$0 + .5(2) = 1$

$0 + .5(3) = 1.5$

≈ 0.58

$\theta_1 = 0, \theta_0 = 0$

$h(\theta) = \theta_0 + \theta_1 x_1) \longrightarrow$

$0 + 0 = 0$

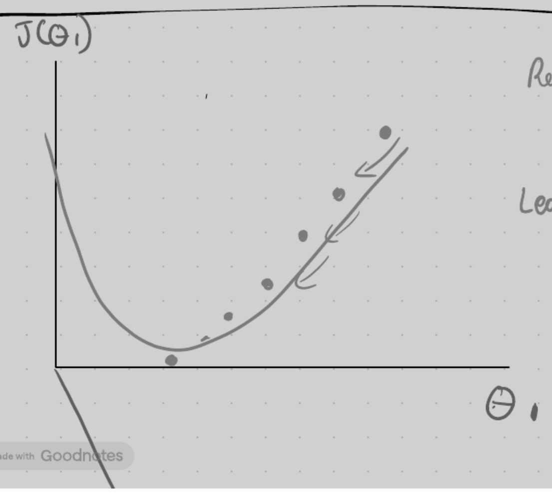
$0 + 0 = 0$

$0 + 0 = 0$

$$J(\theta_1) = \frac{1}{6} \left[(0-1)^2 + (0-2)^2 + (0-3)^2 \right]$$

$J(\theta_1) = \approx 2.3$

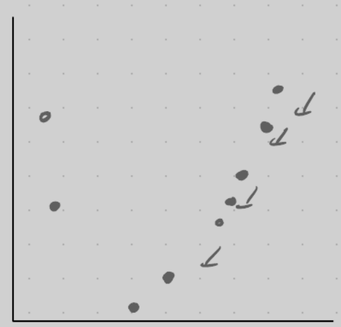
$J(\theta_1, \theta_0)$



Repeat convergence Theorem

Learning Rate

α



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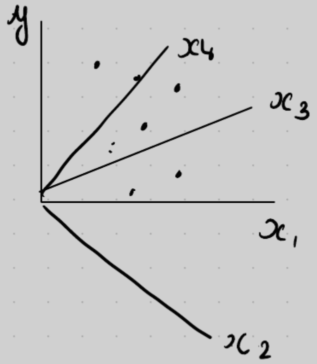
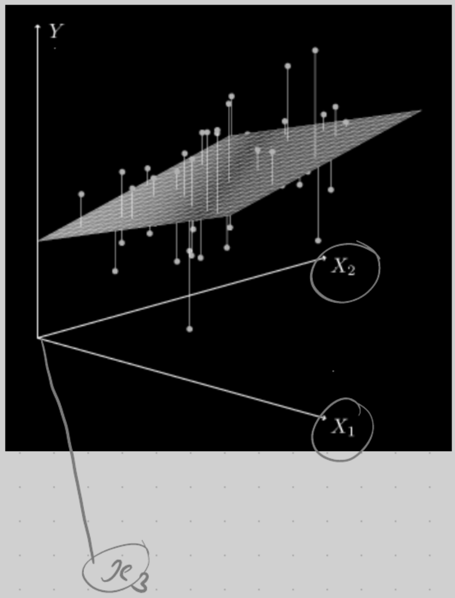
$y = mx + b$

$h(\theta) = \theta_0 + \theta_1 x_1$

$h(\theta_i) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3$

$- - - \theta_n x_n$



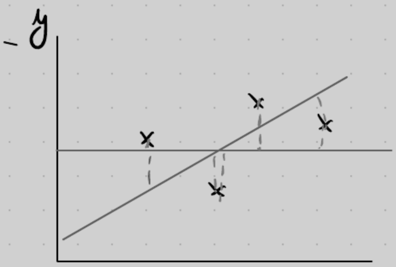


Performance matrices

R^2 Adjusted R^2

$h(\theta)$

$$R^2 = 1 - \frac{\text{Sum of Residuals}}{\text{Sum of total}} \Rightarrow 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$



The Result will be always Positive

The result will be in Decimal

$.86 = 86\%$

Location Age Salary Sleep

$R^2 = 81\%$

$R^2 = 88\%$

$R^2 = 89\%$ ← Model

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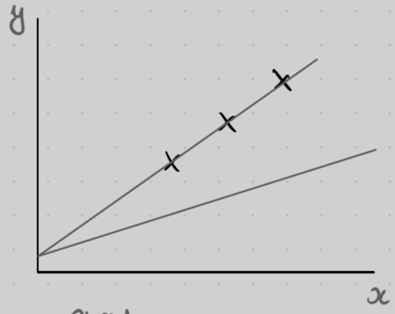
Adjusted R^2

$$\Rightarrow 1 - \frac{(1 - R^2)(n - 1)}{N - \underline{P} - 1}$$

$= n$ = no. of Rows
P = features

Location Age — Salary $\boxed{\cancel{\text{Sleep}}}$

R^2	AR^2	
81%	79%	— $P = 1$
88%	89%	← $P = 2$
⇑ 89%	86% ⇓	



$\underline{J(\theta_0, \theta_1) = \underline{0}}$

Overfitting

{ low bias }

training → model Perform better
Testing → model Not Performed Good
(high variance)

Underfitting

(high bias }

training → model Perform is bad
Testing → model Not Performed Good
(high variance)

M_1	M_2	M_3
Training → 92%	Training → 92%	Training - 68%
Testing → 64%	Testing → 90%	Testing - 64%
⇓	⇓	⇓
Overfitting	Perfet Model	Underfitting
	(low bias)	
	(low variance)	
	⇓	

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Ridge and lasso Regression

cost function

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} \left[(h_\theta x^{(i)}) - y^{(i)})^2 \right] \lambda (\text{slope})^2$$

$= \emptyset \rightarrow$

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